

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1301

Viking Energy Of Northumberland 138 kV
September 18, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1301. The Interconnection Customer proposes a 242.2 MW solar photovoltaic facility interconnecting at the Viking Energy Of Northumberland 138 kV substation in the Virginia area. The request was received on September 27, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 339 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$42,300,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Meyer Rd to Viking Energy Of Northumberland 138 kV line, the Kanaka to Youngcane 138 kV loads to 138.5% of its 156 MVA rating. Project DFAX: 100.0%.

The Viking Energy Of Northumberland to Kanaka 138 kV is overloaded to 119.3% of its 187 MVA emergency rating for the loss of the Meyer Rd Substation to Viking Energy Of Northumberland 138 kV. The project contributes 100.0% DFAX (242.2 MW) to this overload.

Viking Energy Of Northumberland to Meyer Rd 138 kV L: 111.1% loading (201 MVA) following the outage of the Viking Energy Of Northumberland to Kanaka Substation 138 kV; DFAX = 100.0%.

Bobville to Avra Vly 138 kV: 103.7% loading (373 MVA) following the outage of the McCartney to Yarnen 345 kV line; DFAX = 15.7%.

The Foothills to Youngcane Substation 138 kV line is overloaded to 103.3% of its 197 MVA emergency rating for the loss of the Viking Energy Of Northumberland to Meyer Rd 138 kV. The project contributes 100.0% DFAX (242.2 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kanaka to Youngcane 138 kV	Rebuild line with higher-capacity conductor	\$8,684,000	100% (\$8,684,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Viking Energy Of Northumberland to Kanaka 138 kV	Rebuild line with higher-capacity conductor	\$14,707,000	100% (\$14,707,000)
Meyr Rd to Viking Energy O Northumberland 138 kV line	Rebuild line with higher-capacity conductor	\$12,955,000	100% (\$12,955,000)
Bobville to Avra Vly 138 kV	Reconductor line	\$4,905,000	23% (\$1,108,000)
Foothills to Youngcane Substation 138 kV line	Reconductor line	\$4,846,000	100% (\$4,846,000)
	Total allocated to project		\$42,300,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.